Blackmon Lab

Evolutionary Biology

The Role of Polyploidy on Orchidaceae Geographical Range

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Background

Polyploidy, or whole genome duplication, is common in plants and has long been thought to provide increased genetic variation and evolutionary flexibility. In Orchidaceae, a family renowned for its remarkable diversity and global reach, polyploidy may allow species to adapt to new environments and persist across a range of climates. By increasing the number of gene copies, polyploidy can buffer populations against harmful mutations and fuel evolutionary innovation, potentially enabling polyploid orchids to expand their geographic ranges. This correlation between polyploidy and increased geographic range has been observed in families of grasses, but has never been explored in orchid species. Using known and previously collected occurrence data from GBIF (Global Biodiversity Information Facility), maximum and minimum latitudes were collected to find the latitude range that various species occupied. Using known haploid chromosome numbers of multiple Orchidaceae species, the ploidy level of each species was calculated and then sorted into polyploid and diploid. This project aims to understand whether polyploidy allows orchid species to occupy a larger geographical range, as seen in latitude ranges.

Study Design & AI

This study used phylogenetic comparative methods to compare polyploidy and latitudinal ranges of multiple species in the Orchidaceae family. We used phylogenetic generalized least squares (PGLS) to test whether geographical range size differs between diploid and polyploid Orchidaceae species while accounting for phylogenetic non-independence. We analyzed 12 species with complete ploidy and latitudinal range data. AI was used to assist in data collection, data organization, statistical analysis, and visualization of data and results. The Large Language Model (LLM) Claude.AI was primarily used for code development and large data downloads. All coding was performed using R on R Studio, an open source software.

PGLS Summary Statistics				
	Estimate	Standard Error	t value	Pr(>|t|)
Intercept	9.3988	8.7359	1.0759	0.30726
polyploid true	27.1275	9.2225	2.9415	0.01475*
Multiple R-squared: 0.4639	Adjusted R-squared: 0.4103			
F-statistic: 8.652	p-value: 0.01475			

Table 1: The summary statistics from the conducted PGLS analysis

Data & Results

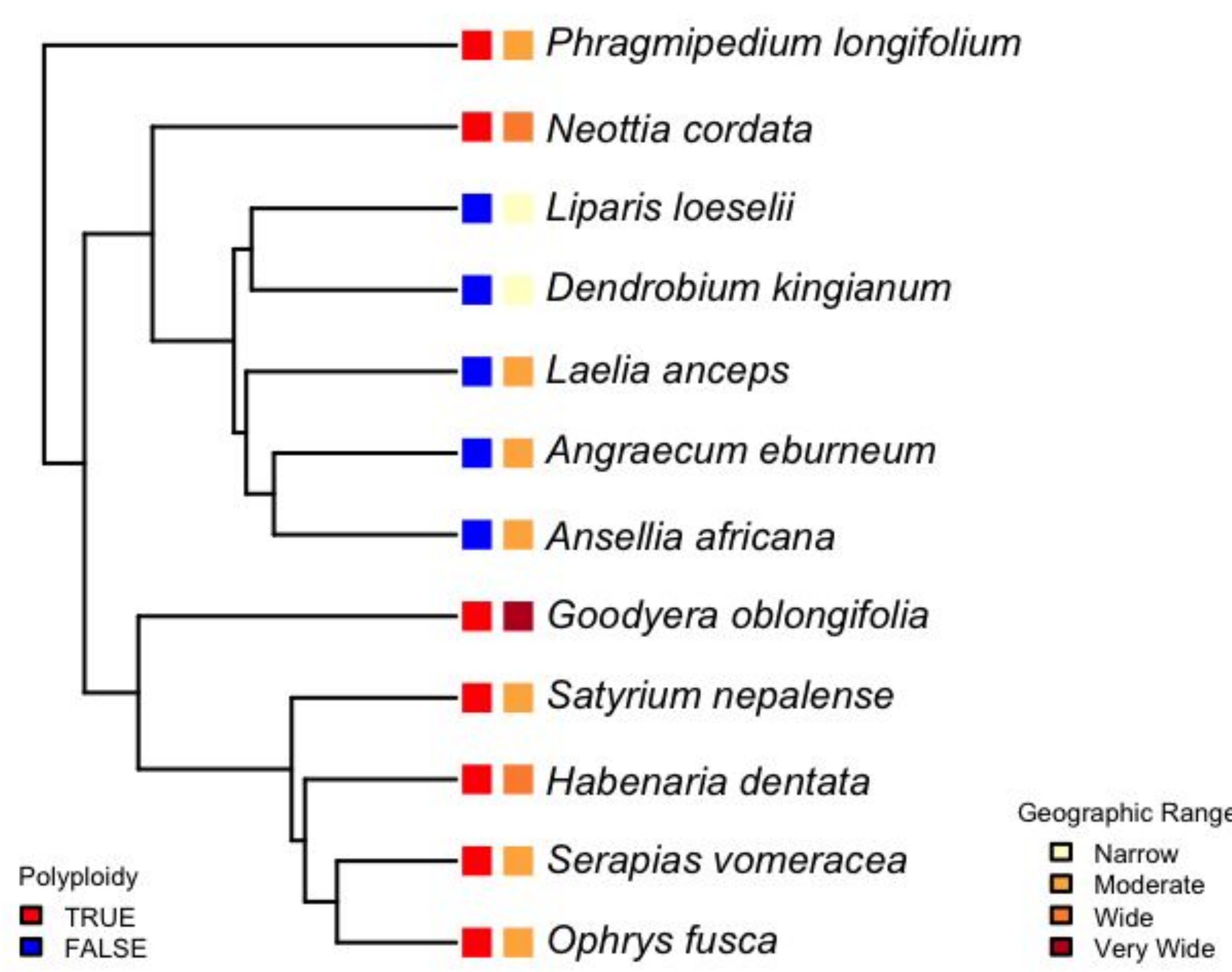
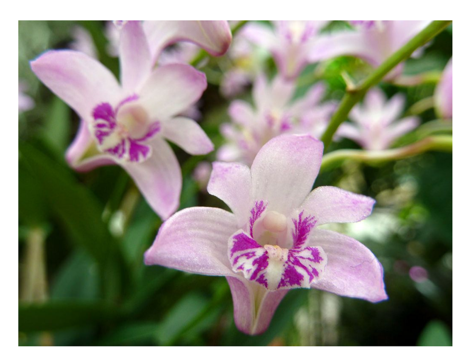
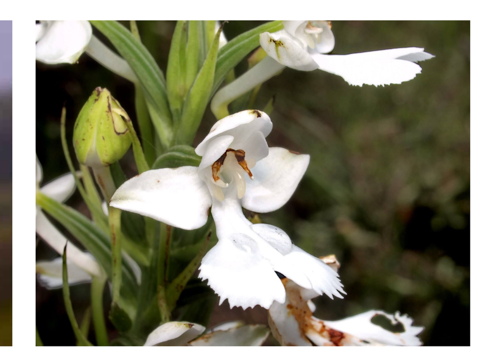


Figure 1: Orchidaceae phylogenetic tree visually depicting the variance in geographic range, as characterized by latitudinal range, per species with their ploidy level.





Across the 12 Orchidaceae species analyzed, polyploid species exhibited substantially larger latitudinal ranges than diploid species. PGLS analysis revealed a significant positive association between polyploidy and geographic range size when accounting for phylogenetic relatedness. Polyploid species had, on average, 27.1° greater latitudinal range than diploid species (Estimate = 27.13, SE = 9.22, t = 2.94, p = 0.0148). The model explained a moderate proportion of variation in range size ($R^2 = 0.46$), indicating that ploidy level is an important predictor of geographic distribution in this dataset.

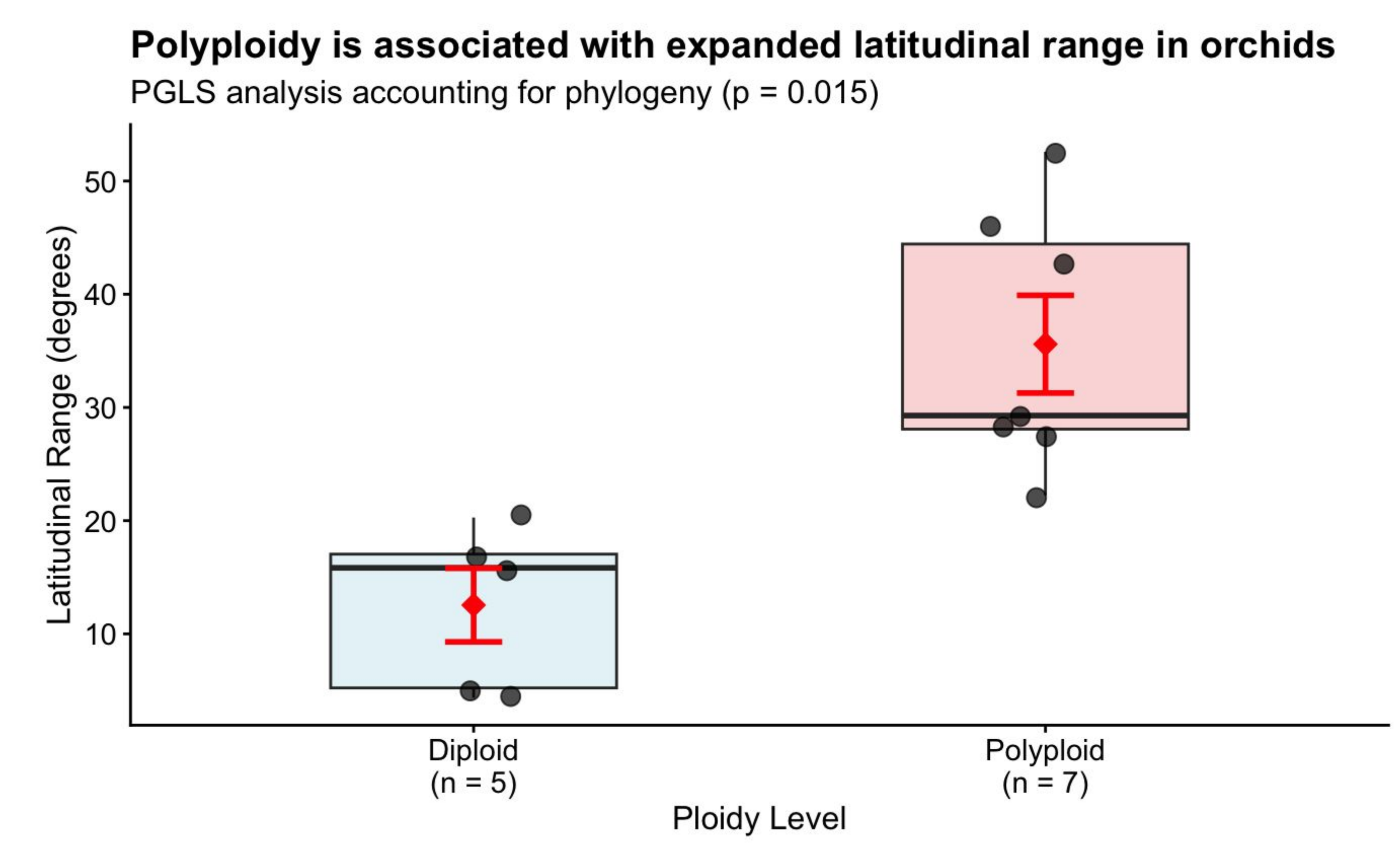


Figure 2: A paired boxplot depicts the collected data separated by ploidy level.

Discussion

Our findings provide preliminary evidence that polyploidy is associated with expanded geographic ranges in Orchidaceae, mirroring patterns previously documented in other plant families such as Poaceae. The significant positive effect of polyploidy on latitudinal range suggests that whole-genome duplication may enhance ecological tolerance or adaptive capacity, enabling polyploid orchids to persist across broader climatic gradients. Several mechanisms may underlie this pattern. Polyploidy increases gene copy number, which can buffer against deleterious mutations and promote functional divergence, potentially facilitating adaptation to new or extreme environments. In orchids, one of the most diverse and ecologically versatile plant families, these genomic advantages may translate into greater climatic resilience and wider dispersal potential, and may even explain orchids' rapid speciation. However, this study is limited by the small sample size and the availability of chromosome count data and occurrence data, which restricts the generalizability of the results. Additionally, latitudinal range is only one dimension of geographic distribution; future work incorporating climatic niche modeling, elevation ranges, or bioclimatic variables could provide a more nuanced understanding of how polyploidy shapes orchid biogeography. Despite these limitations, our results highlight a previously unexplored relationship between genome duplication and geographic range expansion in Orchidaceae. This study contributes to a growing body of evidence that polyploidy plays a meaningful role in plant diversification and ecological success, and it opens the door for broader comparative analyses across the family's thousands of species.

Conclusions & Acknowledgements

In conclusion, there is a relationship between polyploidy and increased geographical range in Orchidaceae, however there is much more work to be done to establish a strong relationship across the family.

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